

Real-Time Data Analyst Coding Challenge

Role: Junior Data Analyst

Objective:

You are working as a Junior Data Analyst in different industries. Your task is to:

1. **Clean the raw dataset**
2. **Perform exploratory data analysis (EDA)**
3. **Visualize the data**
4. **Find business insights**
5. **Write storytelling insights**
6. **Present recommendations**
7. **Create a final dashboard/report**

Each scenario below represents a **real-world business problem** faced by companies today.

E-Commerce Sales Analysis

Scenario

An online retail company is facing declining profits despite increasing sales.

Dataset Contains

- Order ID
- Customer ID
- Product Category
- Discount
- Sales
- Profit
- Region
- Shipping Cost
- Delivery Time

Problem Statement

Identify why profits are decreasing and recommend improvement strategies.

Tasks

- Remove duplicate orders
- Handle missing shipping values
- Analyze discount vs profit relationship
- Find high-loss categories
- Detect regions with delayed deliveries

Expected Visualizations

- Profit by category
- Sales trend over time
- Discount vs Profit scatter plot
- Regional performance heatmap

Challenge

Find:

- Which products destroy profit?
- Are discounts helping or hurting business?
- Which region performs worst?

Storytelling Requirement

Write a business presentation explaining:

- Why profit dropped
- What caused operational inefficiency
- Recommended actions